Executive Summary

Project Overview

The main objective of this project was to analyze Indian Premier League (IPL) cricket match data and understand team performance, player achievements, match results, and the impact of winning the toss. I used Python libraries such as Pandas, Matplotlib, and Seaborn to explore the data and create visualizations, including bar charts, pie charts, and histograms.

Key Findings

1. Overall Match Analysis

The dataset contains **1,024 IPL matches** with information about teams, players, match locations, toss decisions, and match results. Out of these, 1,003 matches (97.95%) had a normal result, while 21 matches (2.05%) were decided using the Duckworth–Lewis method (D/L).

2. Top Player Performances

The Player of the Match analysis identified AB de Villiers as the leading player, with 25 awards, followed by Chris Gayle with 22 and Rohit Sharma with 19. David Warner received 18 awards, while MS Dhoni received 17. These results highlight players who frequently delivered outstanding performances in IPL matches.

3. Team Performance While Batting First

The analysis examined matches won by teams batting first, using the winning margin in runs. Mumbai Indians recorded the highest number of wins in this category, with 69 victories, closely followed by Chennai Super Kings with 67. Royal Challengers Bangalore ranked third with 53 wins.

Among the 463 matches classified as wins by runs in the notebook, Mumbai Indians accounted for approximately 14.90% and Chennai Super Kings for 14.47%. These findings highlight their strong performances when setting a target.

4. Team Performance While Batting Second

The analysis also examined matches won by teams batting second, using the winning margin in wickets. Kolkata Knight Riders recorded the highest number of wins in this category, with 71, followed by Mumbai Indians with 69 and Chennai Super Kings with 64.

These results suggest that Kolkata Knight Riders performed particularly well in the matches classified as wins by wickets in this dataset.

5. Impact of Winning the Toss

One important observation was the relationship between winning the toss and winning the match. Out of 1,024 matches, 517 were won by the team that also won the toss. This represents approximately **50.49%** of all matches.

The result suggests that winning the toss coincided with winning the match in around half of the recorded matches. However, this alone does not prove that winning the toss provides a significant advantage.

6. Match Locations

The city-wise analysis showed that Mumbai hosted the highest number of matches, with 166, followed by Kolkata with 86 and Delhi with 85. Mumbai accounted for approximately 16.21% of all matches in the dataset.

This chart helps identify the cities that hosted the most IPL matches in the available records.

7. Data Exploration Using Ball-by-Ball Records

The project also explored a second dataset containing 243,817 ball-by-ball records and 23 columns. This dataset includes details such as innings, batting teams, bowling teams, striker names, and extras.

I filtered the data to examine deliveries recorded at ball value 2.4 and explored striker frequencies and no-ball information. This demonstrates how detailed delivery-level data can be used to investigate individual player participation and specific match events.

Conclusion

This project helped me understand how data analysis and visualization can be used to explore IPL match statistics and identify meaningful patterns. The charts made it easier

to compare player achievements, team victories, match locations, and the relationship between toss results and match outcomes.

Overall, the analysis highlights AB de Villiers as the leading Player of the Match award recipient in the dataset, Mumbai Indians as the leading team in wins by runs, and Kolkata Knight Riders as the leading team in wins by wickets. It also shows that teams winning the toss won approximately 50.49% of all recorded matches.

The project strengthened my understanding of data cleaning, data filtering, statistical calculations, and visual storytelling using Python. It demonstrates how raw cricket data can be converted into clear insights that help audiences understand team and player performance.

My recommendation

Batting-first and batting-second percentages: Calculate the win percentage using the correct total for each category, rather than comparing raw counts alone.
Ball-by-ball analysis: In the later cells, variables named `srh` and `rcb` are both filtered by `innings == 2`, not by the actual SRH or RCB team name. The resulting player counts therefore do not represent those teams individually. Filter by `batting_team` as well.
Toss analysis: The 50.49% figure is the percentage of all 1,024 records where the toss winner and match winner are the same. For a more precise analysis, verify unique matches and exclude matches without a decisive winner where appropriate.